2019秋高数上期末(回忆版)

1. Determine whether the following statements are true or false? No justification is necessary.

(1) If $k>0$, then $\ln^{100}x < x^{0.0001} < 2^{kx}$ for sufficiently large x.

(2) If f is continuous on R, then $\int_0^a f(a-x)\,dx = \int_0^a f(x)\,dx$.

(3) If the graph of a differentiable function $f(x)$ is concave up on an open interval (a,b), then $f(x)$ has a local minimum value at a point $c \in (a,b)$ if and only if $f'(c)=0$.

(4) If $|f(x)|$ is continuous at $x=a$, then so is $(f(x))^2$.

(5) Suppose that $f(a)=g(a)=0$, that f and g are differentiable on an open interval I containing a, and that $g'(x)\neq 0$ on I if $x\neq a$. If $\lim_{x\to a}\frac{f'(x)}{g'(x)}$ does not exist, then neither does $\lim_{x\to a}\frac{f(x)}{g(x)}$.

2. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) If $g(x)$ is one-to-one, and $g(1)=3$, $g(3)=1$, $g'(1)=4$, $g'(3)=28$, then $(g^{-1})'(3)=$

(A) $\frac{1}{4}$ (B) $\frac{1}{28}$ (C) $\frac{1}{3}$ (D) 4.

(2) Let $c>0$. How many real roots are there for the equation $x^3-6x^2+9x+C=0$?

(A) 0 (B) 1 (C) 2 (D) 3

(3) Suppose $\lim_{x\to 0^+} f(x)=a$, $\lim_{x\to 0^-} f(x)=b$, then $\lim_{x\to 0}\left(f(x-\sin x)+2f(x^2+x)\right)=$

(A) $a+2b$ (B) $b+2a$ (C) $3a$ (D) $3b$.

(4) If $f(x)=\frac{\ln|x|}{|x-1|}\sin x$, then the function $f(x)$ has

(A) 1 removable discontinuity and 1 jump discontinuity.

(B) 2 removable discontinuities.

(C) 1 removable discontinuity and 1 infinite discontinuity.

(D) 2 jump discontinuities.

3. If the function $f(x)=\begin{cases} a\cdot\sin x, & x\le\frac{\pi}{4} \\ 1+b\cdot\tan x, & \frac{\pi}{4}<x<\frac{\pi}{2} \end{cases}$ is differentiable at $x=\frac{\pi}{4}$, find the values of a and b.

(5) Let $f(x)$ be a continuous function, and a is a nonzero constant. Which of the following function is an odd function?

(A) $\int_a^x\left(\int_0^u t f(t^2)\,dt\right)du$ (B) $\int_0^x\left(\int_a^u f(t^3)\,dt\right)du$

(C) $\int_0^x\left(\int_a^u t f(t^2)\,dt\right)du$ (D) $\int_a^x\left(\int_0^u (f(t))^2\,dt\right)du$.

4. Evaluate the following limits.

(1) $\lim_{x\to 0}\frac{\tan^{-1}x - x}{x\tan^2 x}$ (2) $\lim_{x\to\infty}\frac{(x+100)^{100x}}{x^{100x}}$.

5. Find the area of the region enclosed by the curve $y=|x^2-4|$ and $y=\frac{x^2}{2}+4$.

6. The graph of the equation $x^{\frac{2}{3}}+y^{\frac{2}{3}}=1$ is an astroid. Find the area of the surface generated by revolving the curve about the x-axis.

7. The point $P(a,b)$ lies on the curve $l: (y-x)^3=y+x$, and the slope of the tangent line of l at $P(a,b)$ is 3. Find the values of a and b.

8. Find $f'(2)$ if $f(x)=e^{g(x)}$ and $g(x)=\int_2^{\frac{x^2}{2}} \frac{t}{1+t^4}\,dt$.

9. Evaluate the integrals.

(1) $\int \frac{dx}{\sqrt{1+e^x}}$

(2) $\int \frac{3x+6}{(x-1)^2(x^2+x+1)}\,dx$.

(3) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan^2 x \sec x\,dx$.

(4) $\int_{\frac{1}{2}}^{\frac{3}{2}} \frac{1}{\sqrt{|x-x^2|}}\,dx$.

10. An 1600-L tank is half full of fresh water; i.e., contains 800-L of fresh water. At the time $t=0$, a solution containing 0.0625 kg/L of salt runs into the tank at the rate of 16L/min, and the mixture is pumped out of the tank at the rate of 8L/min. At the time the tank is full, how many kilograms of salt will it contain?

11. $f(x)$ is differentiable, and $f'(x)>0$ on $(0,+\infty)$. Let $F(x)=\int_{\frac{1}{x}}^{1} xf(u)\,du+\int_1^{\frac{1}{x}} \frac{f(u)}{u^2}\,du$.

(1) Identify the open intervals on which $F(x)$ is decreasing and the open intervals on which $F(x)$ is increasing.

(2) Find the open intervals on which the graph of $y=F(x)$ is concave up and the open intervals on which it is concave down.

12. Let g be a function that is differentiable throughout an open interval containing the origin. Suppose g has the following properties:

(i) $g(x+y)=\frac{g(x)+g(y)}{1-g(x)g(y)}$ for all real numbers x, y, and $x+y$ in the domain of g.

(ii) $\lim_{h\to 0} g(h)=0$

(iii) $\lim_{h\to 0} \frac{g(h)}{h}=1$.

Find $g(x)$.